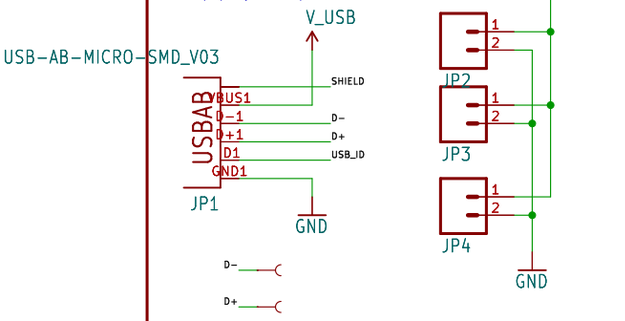
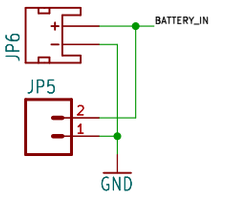
Supply Input



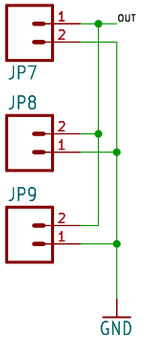
Supply range: 4.35–6.4V

Battery Input

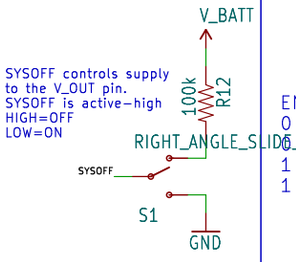


Battery Range: 3.3–4.2V

Supply Output

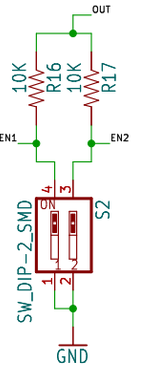


Shutoff Switch



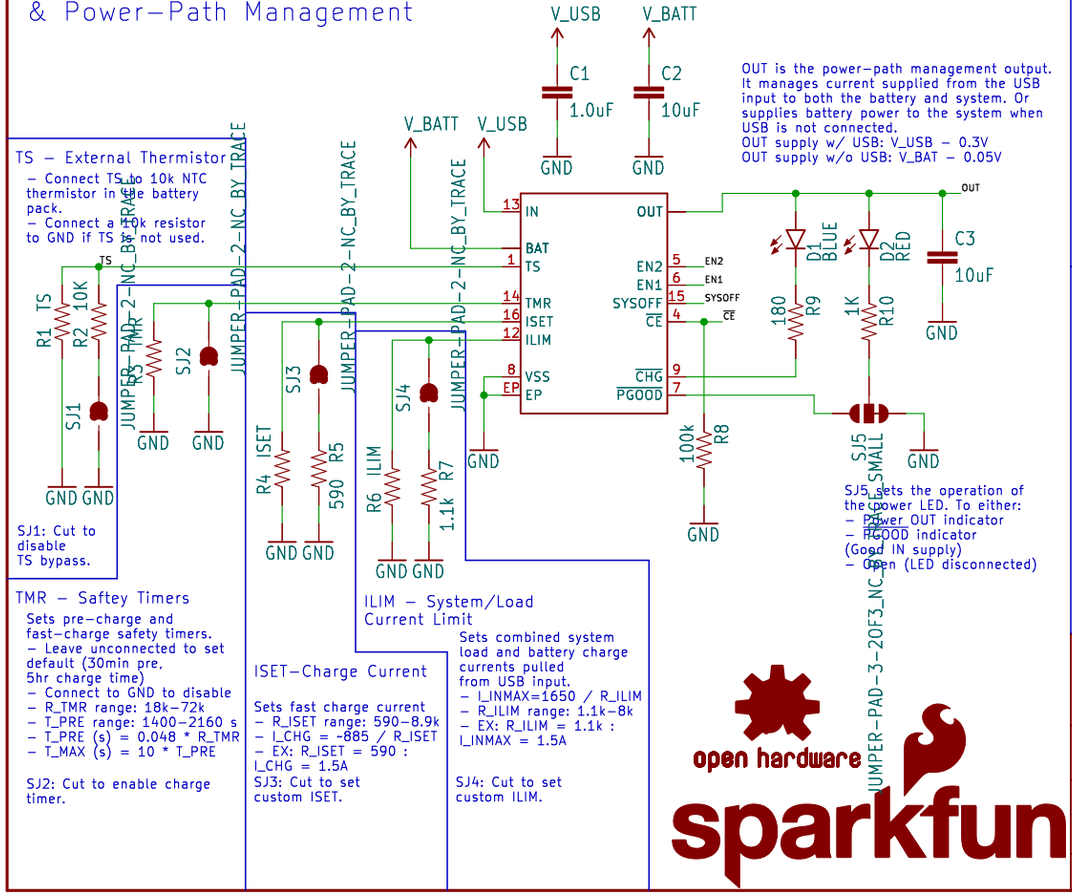
SYSOFF controls supply to the V_OUT pin.
SYSOFF is active-high
HIGH=OFF
LOW=ON

Charge Current Select



EN2	EN1	Current
0	0	100mA
0	1	500mA
1	0	ISET
1	1	Standby

BQ24075 – 1.5A USB-Friendly LiPo Battery Charger & Power-Path Management



OUT is the power-path management output. It manages current supplied from the USB input to both the battery and system. Or supplies battery power to the system when USB is not connected.
OUT supply w/ USB: V_USB – 0.3V
OUT supply w/o USB: V_BAT – 0.05V

TS – External Thermistor

– Connect TS to 10k NTC thermistor in the battery pack.
– Connect a 10k resistor to GND if TS is not used.

SJ1: Cut to disable TS bypass.

TMR – Saftey Timers

Sets pre-charge and fast-charge safety timers.
– Leave unconnected to set default (30min pre, 5hr charge time)
– Connect to GND to disable
– R_TMR range: 18k–72k
– T_PRE range: 1400–2160 s
– T_PRE (s) = 0.048 * R_TMR
– T_MAX (s) = 10 * T_PRE

SJ2: Cut to enable charge timer.

ISET–Charge Current

Sets fast charge current
– R_ISET range: 590–8.9k
– I_CHG = ~885 / R_ISET
– EX: R_ISET = 590 : I_CHG = 1.5A

SJ3: Cut to set custom ISET.

ILIM – System/Load Current Limit

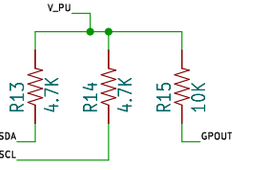
Sets combined system load and battery charge currents pulled from USB input.
– I_INMAX=1650 / R_ILIM
– R_ILIM range: 1.1k–8k
– EX: R_ILIM = 1.1k : I_INMAX = 1.5A

SJ4: Cut to set custom ILIM.

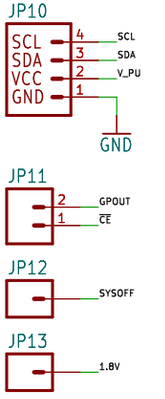
SJ5: Sets the operation of the Power LED. To either:
– Power OUT indicator
– PGOOD indicator (Good IN supply)
– Open (LED disconnected)

I2C Pull-Ups

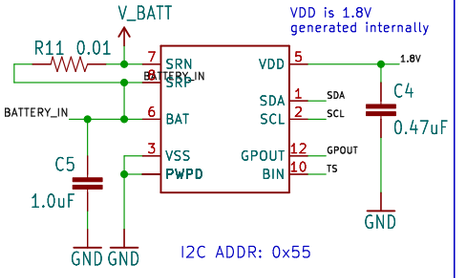
V_PU is an externally supplied voltage to set the pull-up resistor HIGH voltage.
Allowable range: 1.62–6V



Headers



BQ27441-G1 Fuel Gauge



VDD is 1.8V generated internally

I2C ADDR: 0x55

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Design by: Jim Lindblom	REV: v10
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